Title: Student Course Management System

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➢ Introduction:

This project implements a console-based **Student Course Management System** in C++.
The system allows users to manage students and their course enrollments efficiently.
It demonstrates **Object-Oriented Programming**, **STL usage**, and basic C++ concepts.

➢ Detailed Project Description:

- **Objective:** To develop a C++ application that manages students and their enrolled courses.
- **Features Implemented:**
  - Add/Remove/Search students
  - Display all students
  - Enroll students in courses
  - Show student courses
  - Sort students by GPA
- **Limitations:** Data is stored only in memory (no file storage implemented).

➢ Project Beneficiaries:

- Students can track enrolled courses and GPA.
- Teachers can manage student lists and performance.
- Administrators can sort and search student data easily.

➢ Detailed Analysis:

- **Classes:**
  - `Person` (base class) → stores `name`
  - `Student` (derived from `Person`) → stores `id`, `gpa`, `courses`
- **Data Structures:**
  - `vector<Student>` to store student objects
  - `vector<string>` inside Student for courses
- **Algorithms Used:**

- `sort()` to sort students by GPA
- Linear search to find students by ID

➢ Task Breakdown Structure:

Task	Description
Design classes	Create `Person` and `Student` classes with required attributes and methods
Implement menu	Menu-driven interface to perform actions
Add/Delete/Search	Functions to manage student list
Enroll in the course	Add courses to a student, prevent duplicates
Display & Sort	Display all students and sort by GPA

➢ Time Plan

Day	Task
Day 1	Design classes (`Person` & `Student`) + UML diagram
Day 2	Implement classes (`Student.cpp`) and basic methods
Day 3	Implement main menu + Add/Remove/Search/Display students
Day 4	Implement course enrollment + Show courses
Day 5	Implement sort by GPA + Debug all functions
Day 6	Test program + Prepare sample outputs
Day 7	Complete report + README + Final review

➢ System Architecture and Design

- **UML Diagram:**

- **Design Overview:**

  - Main menu interacts with `vector<Student>`
  - Each Student stores courses in a vector
  - Methods ensure encapsulation and prevent duplicate IDs or courses

➢ Testing Scenarios and Results

Test Case	Input	Expected Output	Result
Add Student	ID=1, Name=Karen, GPA=3.5	Student added	Pass
Enroll Course	ID=1, Course=Math	Course added	Pass
Duplicate ID	ID=1	Error message	Pass
Show Courses	ID=1	Display course list	Pass
Sort Students	Multiple students with different GPA	Sorted list	Pass

➢ End-User Guide

- Compile the program using any C++ compiler (Visual Studio, g++, etc.)
- Run the program.
- Use the menu to perform tasks:
  - `1` → Add Student
  - `5` → Enroll Student in Course
  - `6` → Show Courses
  - `7` → Sort by GPA
- Exit with `8`.

➢ Conclusion:

This project demonstrates a functional **Student Course Management System** using C++.
It applies OOP concepts, STL containers, and menu-driven interaction. The system is extendable for future features such as **file saving/loading**.